We will cover the following six topics in the Week 5 lectures:

1. Lagrange multipliers — the two-variable case.
2. Lagrange multipliers — the three-variable case. end of differential calculus
3. Double integrals over a closed standard rectangle in $\mathbb{R}^2$.
4. Double integrals over a general bounded region in $\mathbb{R}^2$.
5. Iterated double integrals.
6. Double integrals over a Type I/Type II region in $\mathbb{R}^2$.

f - real valued function of two real variables

① f_x and f_y continuous @ $p=(a,b)$ $\not\Rightarrow$ f is totally differentiable @ p.

② f_x and f_y continuous @ around p $\Rightarrow$ f is totally differentiable @ p.

③ f is continuous @ interior point p in Dom(f) $\not\Rightarrow$ f is totally differentiable @ p.

④ f is totally differentiable @ p $\Rightarrow$ f is cont. @ p max/mi

In this section, we shall solve the following optimization problem:

1. **Ingredients:**
   - Two real-valued functions of two real variables, f and g.
   - A real number k.
2. **Constraint set:** The set of points $\mathbf{x} \in \mathrm{Dom}(f) \cap \mathrm{Dom}(g)$ such that $g(\mathbf{x}) = k$.
   Restriction on x
3. **Objective:** Maximize/minimize $f(\mathbf{x})$, for points $\mathbf{x}$ in the constraint set.

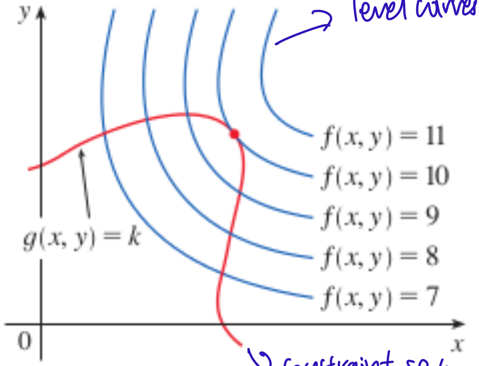
We shall use the Method of Lagrange Multipliers to solve this optimization problem.

Consider the graph surface of f in $\mathbb{R}^3$. Two real variables $z=f(x,y)$

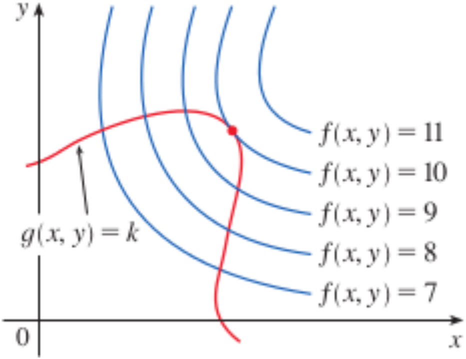
Question: What is the highest point on this surface when we restrict to only points in the constraint set?

Key: View the constraint set as the level curve of g with value k.

level curves of f



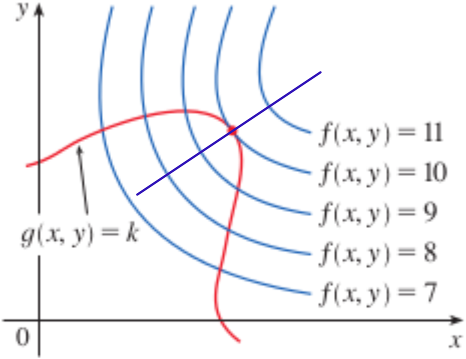
constraint set



New question: Which point on the red curve produces the highest value of f?

Answer: The little red dot (not Singapore, lah).

Conclusion: The maximum value of f <u>subject to the constraint</u> is therefore 10.



Notice how the level curve of f with value 10 and the level curve of g with value k just touch each other at the little red dot?

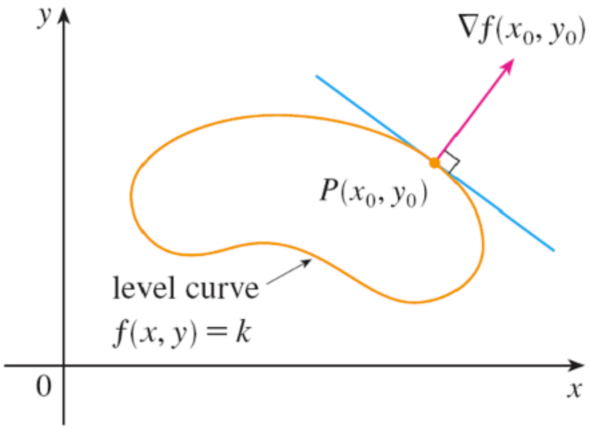
→also share normal line

These two curves thus share a tangent line at the little red dot!

If these two curves had crossed at the little red dot, then there would be another point on the red curve that is situated on a higher level curve of f.

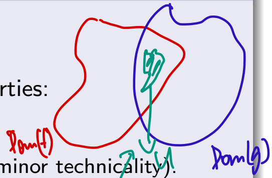
These two curves therefore also share a normal line at the little red dot.

Observation: The gradient of f and the gradient of g are parallel at the little red dot.



Lagrange Multipliers — the Two-Variable Case

Theorem (The Lagrange-Multiplier Theorem for the Two-Variable Case)



- Let f and g be real-valued functions of two real variables.
- Let $k \in \mathbb{R}$. → determines the constraint set
- Let U be an open set in $\mathbb{R}^2$ with the following three properties:
  - $U \subseteq \mathrm{Dom}(f) \cap \mathrm{Dom}(g)$.
  - ∇f and ∇g are defined and continuous throughout U.
  - For all $\mathbf{x} \in U$, we have $g(\mathbf{x}) = k \implies (\nabla g)(\mathbf{x}) \neq (0, 0)$ (a minor technicality).
- Consider the following optimization problem:
  - **Constraint set:** The set of points $\mathbf{x} \in U$ such that $g(\mathbf{x}) = k$.
  - **Objective:** Maximize/minimize $f(\mathbf{x})$, for points $\mathbf{x}$ in the constraint set.
- Suppose that this optimization problem has a solution $\mathbf{p}$.

Then there exists a real number λ, called a "Lagrange multiplier", such that

$$\boxed{(\nabla f)(\mathbf{p}) = \lambda \cdot (\nabla g)(\mathbf{p}).}$$

Caution: This theorem does not guarantee that the stated optimization problem has a solution. It only tells us where to look for a solution, assuming that it exists.

Method for solving a Lagrange-multiplier problem in the two-variable case

(1) Solve the following (often non-linear) system of equations for $\mathbf{p} \in U$ and $\lambda \in \mathbb{R}$:

$$\begin{cases} g(\mathbf{p}) = k \\ (\nabla f)(\mathbf{p}) = \lambda \cdot (\nabla g)(\mathbf{p}) \end{cases} \iff \begin{cases} g(\mathbf{p}) = k \\ f_x(\mathbf{p}) = \lambda \cdot g_x(\mathbf{p}) \\ f_y(\mathbf{p}) = \lambda \cdot g_y(\mathbf{p}) \end{cases}.$$

(2) Find the values of f at all the points $\mathbf{p} \in U$ found in Step 1 (λ is discarded).

Case 1: The constraint set is bounded and closed in $\mathbb{R}^2$.

By the Extreme Value Theorem, the following two statements hold:

(a) The largest value found is the maximum value of f subject to the constraint.

(b) The smallest value found is the minimum value of f subject to the constraint.

Case 2: The constraint set is not bounded or not closed in $\mathbb{R}^2$.

arithmetic mean

geometric mean

Statements (a) and (b) in Case 1 may not hold. Other methods (e.g., the AM-GM Inequality) must be used in conjunction with this step to yield a definitive answer.

$\sqrt{xy} \leq \frac{x+y}{2}$, $\sqrt[3]{xyz} \leq \frac{x+y+z}{3}$, $x,y,z \geq 0$

Example

Define $f : \mathbb{R}^2 \to \mathbb{R}$ by $f(x, y) \stackrel{\text{df}}{=} x^2 + 2y^2$.
Problem: Maximize/minimize f on the unit circle C.

Solution

- Define $g : \mathbb{R}^2 \to \mathbb{R}$ by $g(x, y) \stackrel{\text{df}}{=} x^2 + y^2$. $x^2+y^2=1 \Rightarrow$ constraint
- Then $C = \{(x, y) \in \mathbb{R}^2 \mid g(x, y) = 1\}$, which is the level curve of g with value 1.
- For all $x, y \in \mathbb{R}$, we have

$$(\nabla f)(x, y) = (f_x(x, y), f_y(x, y)) = (2x, 4y),$$
$$(\nabla g)(x, y) = (g_x(x, y), g_y(x, y)) = (2x, 2y).$$

- For $(x, y) \in C$, it thus necessarily holds that $(\nabla g)(x, y) \neq (0, 0)$.

↳ most cases will be true so don't rlly need to check.

- Given $(x, y) \in \mathbb{R}^2$ and $\lambda \in \mathbb{R}$, observe that

$$\begin{cases} g(x,y) = 1 \\ (\nabla f)(x,y) = \lambda \cdot (\nabla g)(x,y) \end{cases} \Rightarrow (2x, 4y) = \lambda(2x, 2y)$$

$$\iff \begin{cases} x^2 + y^2 = 1 \\ 2x = 2\lambda x & \Rightarrow 2x(\lambda - 1) = 0 \\ 4y = 2\lambda y & \Rightarrow 2y(\lambda - 2) = 0 \end{cases}$$

$$\iff \begin{cases} x^2 + y^2 = 1 \\ x = 0 \text{ or } \lambda = 1 \\ y = 0 \text{ or } \lambda = 2 \end{cases}$$

$$\iff \begin{cases} x^2 + y^2 = 1 \\ (x = 0 \text{ and } \lambda = 2) \text{ or } (y = 0 \text{ and } \lambda = 1) \end{cases}$$

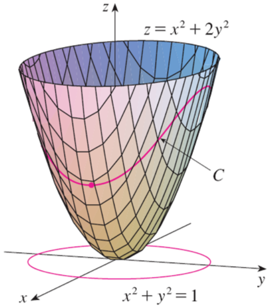
$$\iff \begin{cases} (x,y) = (0, \pm 1) \text{ and } \cancel{\lambda = 2} \\ (x,y) = (\pm 1, 0) \text{ and } \cancel{\lambda = 1} \end{cases}$$

we don't need λ anymore lol

- Compare the values of f at the points $(x, y) \in \mathbb{R}^2$ just found:

(x, y)	$f(x, y)$
$(0, \pm 1)$	2
$(\pm 1, 0)$	1

- As C is bounded and closed in $\mathbb{R}^2$, we can make the following conclusions:
  - The maximum value of f subject to the constraint is 2.
  - The minimum value of f subject to the constraint is 1.



Theorem (The Lagrange-Multiplier Theorem for the Three-Variable Case)

- Let f and g be real-valued functions of three real variables.
- Let $k \in \mathbb{R}$.
- Let U be an open set in $\mathbb{R}^3$ with the following three properties:
  - $U \subseteq \mathrm{Dom}(f) \cap \mathrm{Dom}(g)$.
  - ∇f and ∇g are defined and continuous throughout U.
  - For all $\mathbf{x} \in U$, we have $g(\mathbf{x}) = k \implies (\nabla g)(\mathbf{x}) \neq (0,0,0)$ (a minor technicality). should hold most of the time
- Consider the following optimization problem:
  - **Constraint set:** The set of points $\mathbf{x} \in U$ such that $g(\mathbf{x}) = k$.
  - **Objective:** Maximize/minimize $f(\mathbf{x})$, for points $\mathbf{x}$ in the constraint set.
- Suppose that this optimization problem has a solution $\mathbf{p}$.

Then there exists a real number λ, called a "Lagrange multiplier", such that

$$(\nabla f)(\mathbf{p}) = \lambda \cdot (\nabla g)(\mathbf{p}).$$

Note: The constraint set $\{\mathbf{x} \in U \mid g(\mathbf{x}) = k\}$ is now a surface in $\mathbb{R}^3$.

Caution: This theorem does not guarantee that the stated optimization problem has a solution. It only tells us where to look for a solution, assuming that it exists.

Method for solving a Lagrange-multiplier problem in the three-variable case

(1) Solve the following (often non-linear) system of equations for $\mathbf{p} \in U$ and $\lambda \in \mathbb{R}$:

$$\begin{cases} g(\mathbf{p}) = k \\ (\nabla f)(\mathbf{p}) = \lambda \cdot (\nabla g)(\mathbf{p}) \end{cases} \iff \begin{cases} g(\mathbf{p}) = k \\ f_x(\mathbf{p}) = \lambda \cdot g_x(\mathbf{p}) \\ f_y(\mathbf{p}) = \lambda \cdot g_y(\mathbf{p}) \\ f_z(\mathbf{p}) = \lambda \cdot g_z(\mathbf{p}) \end{cases}.$$

(2) Find the values of f at all the points $\mathbf{p} \in U$ found in Step 1 (λ is discarded).

Case 1: The constraint set is bounded <u>and</u> closed in $\mathbb{R}^3$.

By the Extreme Value Theorem, the following two statements hold:

(a) The largest value found is the maximum value of f <u>subject to the constraint</u>.
(b) The smallest value found is the minimum value of f <u>subject to the constraint</u>.

Case 2: The constraint set is not bounded <u>or</u> not closed in $\mathbb{R}^3$.

Statements (a) and (b) in Case 1 may not hold. Other methods (e.g., the AM-GM Inequality) must be used in conjunction with this step to yield a definitive answer.

Example

A rectangular box without a top lid is to be made from 12 square units of cardboard.
area
Problem: Find the maximum volume of such a box.

Solution

- Given a box without a top lid, its length x, its width y, its height z, its volume V, and its surface area A are related as

$$V = xyz \quad \text{and} \quad A = xy + 2xz + 2yz.$$

- Hence, define $V : \mathbb{R}^3_{>0} \to \mathbb{R}$ and $A : \mathbb{R}^3_{>0} \to \mathbb{R}$ by

$x \quad \frac{x}{z} \quad x \quad \frac{x}{z}$

$$V(x, y, z) \overset{\text{df}}{=} xyz \quad \text{and} \quad A(x, y, z) \overset{\text{df}}{=} xy + 2xz + 2yz.$$

- We then wish to solve the following optimization problem.
  - **Constraint set:** The set of points $(x, y, z) \in \mathbb{R}^3_{>0}$ such that $A(x, y, z) = 12$.
  - **Objective:** Maximize $V(x, y, z)$, for points (x, y, z) in the constraint set.
- For all $x, y, z \in \mathbb{R}_{>0}$, we have

$$(\nabla V)(x, y, z) = (V_x(x, y, z), V_y(x, y, z), V_z(x, y, z)) = (yz, xz, xy),$$
$$(\nabla A)(x, y, z) = (A_x(x, y, z), A_y(x, y, z), A_z(x, y, z)) = (y + 2z, x + 2z, 2x + 2y);$$

clearly, $(\nabla A)(x, y, z) \neq (0, 0, 0)$.

- Given $(x, y, z) \in \mathbb{R}_{>0}$ and $\lambda \in \mathbb{R}$, observe that

$$\begin{cases} A(x,y,z) = 12 \\ (\nabla V)(x,y,z) = \lambda \cdot (\nabla A)(x,y,z) \end{cases}$$

$$\iff \begin{cases} xy + 2xz + 2yz = 12 \\ yz = \lambda(y + 2z) & \times x \\ xz = \lambda(x + 2z) & \times y \\ xy = 2\lambda(x + y) & \times z \end{cases}$$

$$\iff \begin{cases} xy + 2xz + 2yz = 12 \\ xyz = \lambda x(y + 2z) \\ xyz = \lambda y(x + 2z) \\ xyz = 2\lambda z(x + y) \end{cases}$$

(As $x, y, z \neq 0$.)

$xyz = \lambda xy + 2\lambda xz$
$-\ xyz = \lambda xy + 2\lambda yz$
$0 = 0 + 2\lambda z(x - y)$

$2\lambda z(x-y) = 0$, $\lambda z \neq 0$

$\lambda(x-y) = 0$, $\lambda \neq 0$

$\therefore x - y = 0$

$x = y$

$2\lambda z x + 2\lambda z y = 2\lambda z x + \lambda x y$

$2\lambda z y = \lambda x y$

$2\lambda z y - \lambda x y = 0$

$\lambda y(2z - x) = 0$

$$\iff \begin{cases} xy + 2xz + 2yz = 12 \\ xyz = \lambda x(y + 2z) \\ x = y \\ x = 2z \end{cases}.$$

(After elimination; also, $\lambda \neq 0$.)

$xy + 2xz + 2yz = 12$

fix $z = 1$

let $y \to 6$

- Hence,

$$x(x) + 2x\left(\frac{x}{2}\right) + 2(x)\left(\frac{x}{2}\right) = 12 \iff 3x^2 = 12 \iff x = 2,$$

$xy + 2x + 2y = 12$

$x(y+2) = 12 - 2y$

$x = \frac{12-2y}{y+2}$

so the only solution of the Lagrange-multiplier problem is

$$(x, y, z, \lambda) = \left(2, 2, 1, \frac{1}{2}\right).$$

all coordinates should be positive

- Discarding $\lambda = \frac{1}{2}$, we obtain $V(2,2,1) = 4$.

not in constraint set → x-coordinate cannot be zero!

- The constraint set is not closed because $(0, 6, 1)$ is one of its limit points in $\mathbb{R}^3$.
- The constraint set is, however, bounded — given (x, y, z) in the set, the AM-GM Inequality yields the following relations:

can approach $(0,6,1)$ from constraint set.

$$\sqrt[3]{(xy)(2xz)(2yz)} \leq \frac{xy + 2xz + 2yz}{3},$$
$$\sqrt[3]{4x^2y^2z^2} \leq \frac{12}{3} = 4,$$
$$xyz \leq 4.$$

$\to 4x^2y^2z^2 \leq 64$

$x^2y^2z^2 \leq 16$

- Therefore, V achieves a maximum value of 4 subject to the constraint.

A Review of Riemann Integration from Single-Variable Calculus

Let us briefly review Riemann integration from single-variable calculus.

Let f be a real-valued function defined on a bounded and closed interval $I = [a, b]$.

Recall that a sampling x^* of I is a function with the following two properties:

- $\mathrm{Dom}(x^*) = \left\{(n, i) \in \mathbb{N}^2 \;\middle|\; 1 \le i \le n\right\}$.
- Given $(n, i) \in \mathrm{Dom}(x^*)$, its output under x^* — often denoted by $x^*_{n,i}$ and called a "sample point" — is a real number in the following sub-interval of I:

ith sub-interval out of n sub-intervals

$$\boxed{\left[a + (i-1)\left(\frac{b-a}{n}\right), a + i\left(\frac{b-a}{n}\right)\right]}.$$

A Riemann integral of f over I is then defined as an $L \in \mathbb{R}$ such that, regardless of the sampling x^* of I chosen, we have

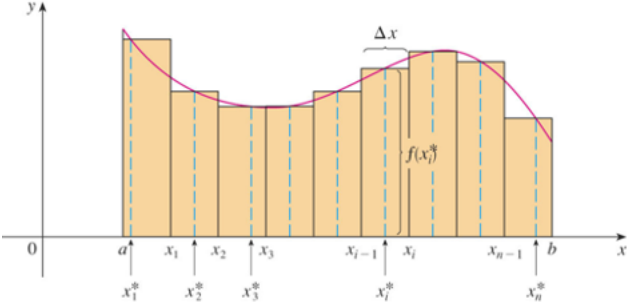
Riemann Sum

$$\lim_{n\to\infty} \sum_{i=1}^{n} f\left(x^*_{n,i}\right) \cdot \underbrace{\left(\frac{b-a}{n}\right)}_{(\Delta x)_n} = L;$$

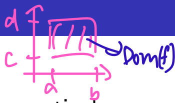
width of sub-interval

if such an $L \in \mathbb{R}$ exists, then it is unique, in which case we denote it by $\displaystyle\int_a^b f(x)\,\mathrm{d}x$.

A visual example of a Riemann sum over a bounded and closed interval.



Double Integrals over a Closed Standard Rectangle in $\mathbb{R}^2$

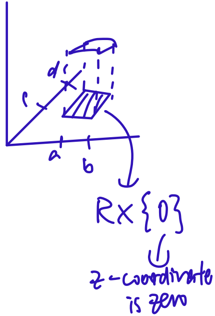


Let $R = [a, b] \times [c, d]$ be a closed standard rectangle in $\mathbb{R}^2$.
Remark: "Standard" means that each side of the rectangle is horizontal or vertical.

Let f be a non-negative (≥ 0) real-valued function defined on R (not necessarily continuous).

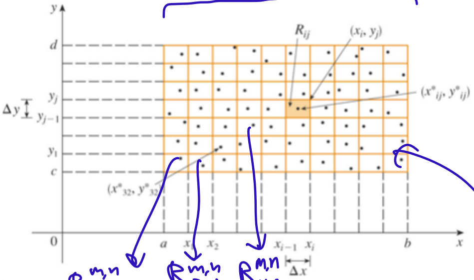
The graph of f is a surface in $\mathbb{R}^3$ lying above the closed rectangular region $R \times \{0\}$ in the $z = 0$ plane. $f(x,y)$

Problem: Define the volume between $R \times \{0\}$ and the graph of f.



Double Integrals over a Closed Standard Rectangle in $\mathbb{R}^2$

m subintervals



n subintervals

m×n subrectangles

$R^{m,n}_{1,1}$ $R^{m,n}_{2,1}$ $R^{m,n}_{4,3}$

formed by the → ith x-subinterval and jth y-subinterval

Given $m, n, i, j \in \mathbb{N}$ that satisfy $1 \leq i \leq m$ and $1 \leq j \leq n$, denote by $R^{m,n}_{i,j}$ the following sub-rectangle of R:

$$\left[a + (i-1)\left(\frac{b-a}{m}\right), a + i\left(\frac{b-a}{m}\right)\right] \times \left[c + (j-1)\left(\frac{d-c}{n}\right), d + j\left(\frac{d-c}{n}\right)\right].$$

A sampling of R is then defined as a function $\mathbf{x}^*$ with the following two properties:

- $\mathrm{Dom}(\mathbf{x}^*) = \{((m,n),(i,j)) \in \mathbb{N}^2 \times \mathbb{N}^2 \mid 1 \leq i \leq m \text{ and } 1 \leq j \leq n\}$.
- Given $((m,n),(i,j)) \in \mathrm{Dom}(\mathbf{x}^*)$, its output under $\mathbf{x}^*$ — denoted by $\mathbf{x}^*_{(m,n),(i,j)}$ and called a "sample point" — is a point in the sub-rectangle $R^{m,n}_{i,j}$ of R.

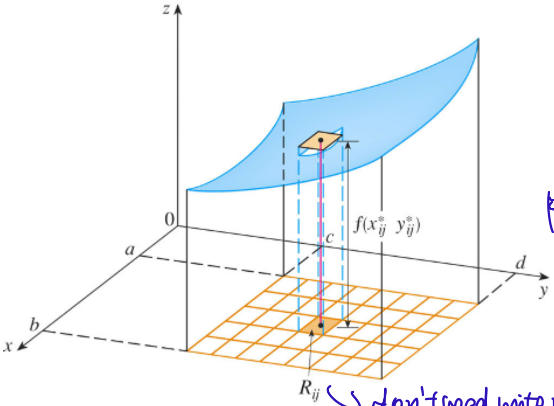
Intuitively, the volume between $R_{i,j}^{m,n} \times \{0\}$ and the graph of f can be approximated as

$$f\left(\mathbf{x}^*_{(m,n),(i,j)}\right) \cdot \operatorname{Area}\left(R_{i,j}^{m,n}\right) = f\left(\mathbf{x}^*_{(m,n),(i,j)}\right) \cdot \underbrace{\left(\frac{b-a}{m}\right)\left(\frac{d-c}{n}\right)}_{(\Delta A)_{m,n}}.$$

(x,y)

height

Base area × height



Fix m=5
n=8

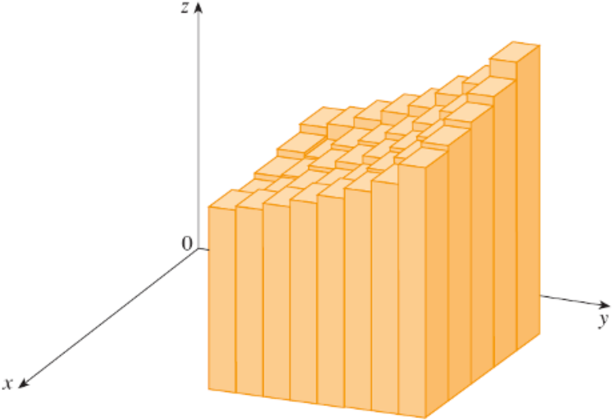
we already fixed it

don't need write m,n

Hence, the volume between $R \times \{0\}$ and the graph of f can be approximated as

$$\sum_{i=1}^{m} \sum_{j=1}^{n} f\left(\mathbf{x}^*_{(m,n),(i,j)}\right) \cdot \underbrace{\left(\frac{b-a}{m}\right)\left(\frac{d-c}{n}\right)}_{(\Delta A)_{m,n}}.$$

Riemann Sum in 2D



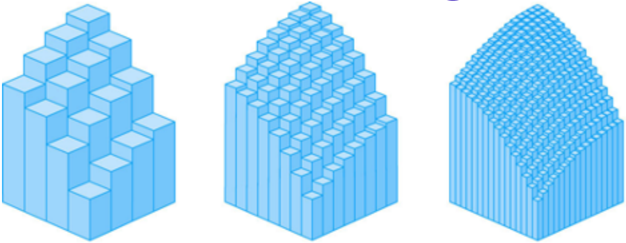
Geometric intuition tells us that as $m, n \to \infty$, this approximation gets better, so let us define a volume between $R \times \{0\}$ and the graph of f as a non-negative real number V such that, regardless of the sampling $\mathbf{x}^*$ of R chosen, we have

$$\lim_{m,n\to\infty} \sum_{i=1}^{m} \sum_{j=1}^{n} f\left(\mathbf{x}^*_{(m,n),(i,j)}\right) \cdot \underbrace{\left(\frac{b-a}{m}\right)\left(\frac{d-c}{n}\right)}_{(\Delta A)_{m,n}} = V;$$

may not exist

if such a non-negative real number V exists, then it is unique.

if it exist, must be unique



(a) $m = n = 4$, $V \approx 41.5$ (b) $m = n = 8$, $V \approx 44.875$ (c) $m = n = 16$, $V \approx 46.46875$

Definition (Double Integrals over a Closed Standard Rectangle in $\mathbb{R}^2$)

- Let $R = [a, b] \times [c, d]$ be a closed standard rectangle in $\mathbb{R}^2$.
- Let f be a real-valued function defined on R (not necessarily continuous).

not necessarily nonnegative

region of integration

A double integral of f over R is then defined as an $L \in \mathbb{R}$ such that, regardless of the sampling $\mathbf{x}^*$ of R chosen, we have

$$\boxed{\lim_{m,n\to\infty} \sum_{i=1}^{m} \sum_{j=1}^{n} f\left(\mathbf{x}^*_{(m,n),(i,j)}\right) \cdot \underbrace{\left(\frac{b-a}{m}\right)\left(\frac{d-c}{n}\right)}_{(\Delta A)_{m,n}}} = L;$$

Riemann sums

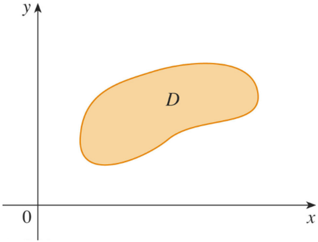
Area

if such an $L \in \mathbb{R}$ exists, then it is unique, in which case we denote it by $\iint_R f(x, y)\, dA$.

double integral

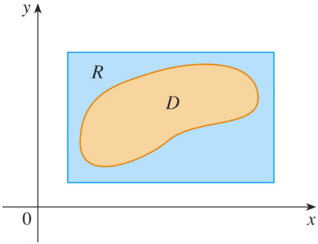
Double Integrals over a General Bounded Region in $\mathbb{R}^2$

Let D be a bounded region in $\mathbb{R}^2$ (not necessarily closed).

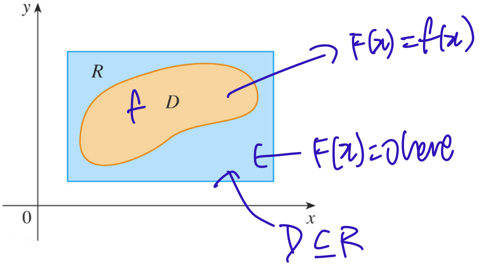


Bounded but not closed.

We can then enclose it within a closed standard rectangle R in $\mathbb{R}^2$ as shown.



Double Integrals over a General Bounded Region in $\mathbb{R}^2$



Let f be a real-valued function defined on D.

Extension by 0.

Define a new function $F : R \to \mathbb{R}$ by

$$F(\mathbf{x}) \stackrel{\text{df}}{=} \begin{cases} f(\mathbf{x}), & \text{if } \mathbf{x} \in D; \\ 0, & \text{if } \mathbf{x} \in R \setminus D. \end{cases}$$

Hence, F, defined on a larger domain, coincides with f on D and is 0 outside of D.

Double Integrals over a General Bounded Region in $\mathbb{R}^2$ — a Definition

Definition (Double Integrals over a General Bounded Region in $\mathbb{R}^2$)

- Let D be a bounded region in $\mathbb{R}^2$ (not necessarily closed).
- Let f be a real-valued function defined on D (not necessarily continuous).

If R is a closed standard rectangle in $\mathbb{R}^2$ such that $D \subseteq R$ and $\displaystyle\iint_R F(x, y)\, \mathrm{d}A$ exists, where $F : R \to \mathbb{R}$ is defined by

1 2

$$F(\mathbf{x}) \stackrel{\text{df}}{=} \begin{cases} f(\mathbf{x}), & \text{if } \mathbf{x} \in D; \\ 0, & \text{if } \mathbf{x} \in R \setminus D, \end{cases}$$

Agrees with f on D
but is Outside it.

then the double integral of f over D, denoted by $\displaystyle\iint_D f(x, y)\, \mathrm{d}A$, is defined as follows:

$$\iint_D f(x, y)\, \mathrm{d}A \stackrel{\text{df}}{=} \iint_R F(x, y)\, \mathrm{d}A;$$

if such a closed standard rectangle R does not exist, then we say that $\displaystyle\iint_D f(x, y)\, \mathrm{d}A$ does not exist (or is not defined).

In the preceding definition, $\displaystyle\iint_D f(x,y)\,\mathrm{d}A$, whenever it is defined, seems to depend on how the enclosing closed standard rectangle R is chosen.

However, this seeming dependence is merely illusory, as the next theorem shows.

Theorem (Consistency of the Definition of General Double Integrals)

- Let D be a bounded region in $\mathbb{R}^2$ (not necessarily closed).
- Let f be a real-valued function defined on D (not necessarily continuous).
- Let R and S be closed standard rectangles in $\mathbb{R}^2$ enclosing D.
- Define $F : R \to \mathbb{R}$ and $G : S \to \mathbb{R}$ by

$$F(\mathbf{x}) \stackrel{\text{df}}{=} \begin{cases} f(\mathbf{x}), & \text{if } \mathbf{x} \in D; \\ 0, & \text{if } \mathbf{x} \in R \setminus D \end{cases} \quad \text{and} \quad G(\mathbf{x}) \stackrel{\text{df}}{=} \begin{cases} f(\mathbf{x}), & \text{if } \mathbf{x} \in D; \\ 0, & \text{if } \mathbf{x} \in S \setminus D. \end{cases}$$

Then $\displaystyle\iint_R F(x,y)\,\mathrm{d}A$ exists if and only if $\displaystyle\iint_S G(x,y)\,\mathrm{d}A$ exists, in which case both of these double integrals are equal.

it's unique!!

Double Integrals over a General Bounded Region in $\mathbb{R}^2$ — Areas

The double integral can be used to give a rather general definition of “area”.

Definition (the Area of a Bounded Region in $\mathbb{R}^2$)

Let D be a bounded region in $\mathbb{R}^2$ (not necessarily closed).

The area of D, denoted by Area(D), is defined as $\displaystyle\iint_D 1 \, dA$ — if it exists.

Not every bounded region in $\mathbb{R}^2$ has an area, for example, $(\mathbb{Q} \cap [0,1]) \times (\mathbb{Q} \cap [0,1])$.

However, most bounded regions in $\mathbb{R}^2$ that you are familiar with have an area, such as disks in $\mathbb{R}^2$ (closed or open) and polygonal regions in $\mathbb{R}^2$ (closed or open).

Note that points and line segments in $\mathbb{R}^2$ have zero area.

In fact, we have the following useful theorem.

not tested but good to know

Theorem (a Necessary and Sufficient Condition for the Existence of an Area)

A bounded region in $\mathbb{R}^2$ has an area if and only if its boundary has zero area.

Double Integrals over a General Bounded Region in $\mathbb{R}^2$ — Existence

We would like to have a result that at least tells us when general double integrals exist.

If you are thinking about continuity, then you are actually on the right track.

Theorem (the Lebesgue Integrability Criterion)

- Let D be a bounded region in $\mathbb{R}^2$ whose area exists.
- Let f be a real-valued function defined on D.

Then $\iint_D f(x, y)\, \mathrm{d}A$ exists if and only if the following two conditions hold:

- f is bounded on D.
- f is continuous at "almost every" point in D.

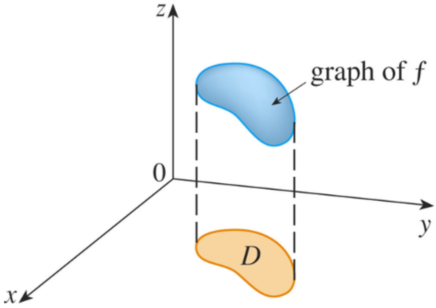
This result guarantees the existence of general double integrals in virtually all problems that we shall ever encounter.

Definition (General Double Integrals as Volumes)

- Let D be a bounded region in $\mathbb{R}^2$ whose area exists.
- Let f be a bounded, continuous, non-negative, real-valued function defined on D.

The volume V between $D \times \{0\}$ and the graph of f is then defined as

$$V \overset{\text{df}}{=} \iint_D f(x, y)\, \mathrm{d}A.$$



Let D be a bounded region in $\mathbb{R}^2$ whose area exists.

A mass-density function on D is just a bounded, continuous, non-negative, real-valued function ρ defined on D.

The total mass m of D is then defined as $m \stackrel{\text{df}}{=} \iint_D \rho(x, y)\, dA$.

$$\text{Mass} = \iint_D \text{Density}(x,y)\, dA$$

we also use double integrals to calculate centre of mass, moment of enertia.

pdf

The joint probability density function of two real-valued random variables, X and Y, is a non-negative real-valued function f defined on $\mathbb{R}^2$ such that

Total probability $\leftarrow$ $$\iint_{\mathbb{R}^2} f(x, y)\, \mathrm{d}A = 1.$$

$\sum$ discrete replaced by $\int$ continuous

not a rectangle!

Remark: As the domain of integration, $\mathbb{R}^2$, is unbounded, this double integral is known as an improper double integral.

$\sum\sum$ discrete replaced by $\iint$ cont

The probability that we can find the value of (X, Y) in a general bounded region D in $\mathbb{R}^2$ is then calculated as

$$\textbf{Prob}((X, Y) \in D) \overset{\text{df}}{=} \underbrace{\iint_D f(x, y)\, \mathrm{d}A}_{\leq 1}. \quad \text{(If it exists.)}$$

Once again, let $R = [a, b] \times [c, d]$ be a closed standard rectangle.

Let f be a real-valued function defined on R.

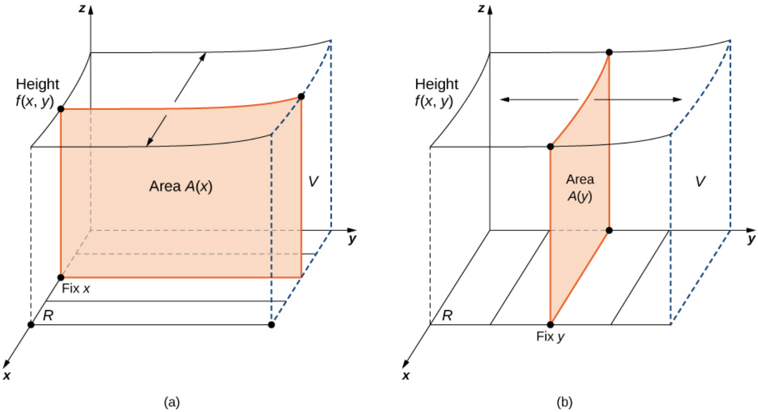
Then the expressions

outer → inner integral   outer → inner integral

$$\int_a^b \left[\int_c^d f(x, y)\, \mathrm{d}y \right] \mathrm{d}x \quad \text{and} \quad \int_c^d \left[\int_a^b f(x, y)\, \mathrm{d}x \right] \mathrm{d}y$$

are known as the iterated double integrals of f over R.

depends on x   depends on y



The iterated double integrals are interpreted according to the diagram above:

$$\int_a^b \left[\int_c^d f(x, y)\, \mathrm{d}y \right] \mathrm{d}x = \int_a^b A(x)\, \mathrm{d}x,$$

$$\int_c^d \left[\int_a^b f(x, y)\, \mathrm{d}x \right] \mathrm{d}y = \int_c^d A(y)\, \mathrm{d}y.$$

Example

Problem: Evaluate the iterated double integral $\displaystyle\int_1^2 \left[\int_0^3 x^2 y \, \mathrm{d}x\right] \mathrm{d}y$.

Solution

- For each $y \in [1, 2]$, we have

$$\int_0^3 x^2 y \, \mathrm{d}x = y \int_0^3 x^2 \, \mathrm{d}x = y \left[\frac{x^3}{3}\right]\Big|_0^3 = y \left[\frac{3^3}{3} - \frac{0^3}{3}\right] = y[9] = 9y.$$

only y left!

- Then

$$\int_1^2 \left[\int_0^3 x^2 y \, \mathrm{d}x\right] \mathrm{d}y = \int_1^2 9y \, \mathrm{d}y = \left[\frac{9y^2}{2}\right]\Big|_1^2 = \frac{9(2)^2}{2} - \frac{9(1)^2}{2} = \frac{27}{2} = 13.5.$$

Example

Problem: Evaluate the iterated double integral $\int_0^3 \left[\int_1^2 x^2 y \; \mathrm{d}y\right] \mathrm{d}x$.

Solution

- For each $x \in [0, 3]$, we have

$$\int_1^2 x^2 y \; \mathrm{d}y = x^2 \int_1^2 y \; \mathrm{d}y = x^2 \left[\frac{y^2}{2}\right]\Bigg|_1^2 = x^2 \left[\frac{2^2}{2} - \frac{1^2}{2}\right] = x^2 \left[\frac{3}{2}\right] = \frac{3x^2}{2}.$$

- Then

$$\int_0^3 \left[\int_1^2 x^2 y \; \mathrm{d}y\right] \mathrm{d}x = \int_0^3 \frac{3x^2}{2} \; \mathrm{d}x = \left[\frac{x^3}{2}\right]\Bigg|_0^3 = \frac{3^3}{2} - \frac{0^3}{2} = \frac{27}{2} = 13.5.$$

In the preceding two examples, we obtained the same answers.

It appears that the order of integration in an iterated double integral does not matter.

In most cases of interest to us, it indeed does not matter, as Guido Fubini verified.

Theorem (Fubini's Theorem)

- Let $R = [a, b] \times [c, d]$ be a closed standard rectangle.
- Let f be a continuous real-valued function defined on R.

Then

$$\int_a^b \left[\int_c^d f(x, y)\, \mathrm{d}y \right] \mathrm{d}x = \iint_R f(x, y)\, \mathrm{d}A = \int_c^d \left[\int_a^b f(x, y)\, \mathrm{d}x \right] \mathrm{d}y.$$

Fubini's Theorem therefore assures us that the double integral of a continuous function defined on a closed standard rectangle can be evaluated as an iterated double integral, in two different ways.

☆ we should try to do order where integration is easier.

Iterated Double Integrals — Example 3

Example

Let $R \overset{\text{df}}{=} \overset{x}{[1, 2]} \times \overset{y}{[0, \pi]}$.

Problem: Evaluate the double integral $\displaystyle\iint_R y \sin(xy)\, \mathrm{d}A$ in two different ways.

Solution

- By Fubini's Theorem, we have

$$\begin{aligned}
\iint_R y \sin(xy)\, \mathrm{d}A &= \int_0^{\pi} \left[\int_1^2 y \sin(xy)\, \mathrm{d}x \right] \mathrm{d}y \\
&= \int_0^{\pi} [-\cos(xy)]\big|_{x=1}^{x=2}\, \mathrm{d}y \\
&= \int_0^{\pi} [\cos(y) - \cos(2y)]\, \mathrm{d}y \\
&= \left[\sin(y) - \frac{1}{2}\sin(2y) \right] \Bigg|_0^{\pi} \\
&= \left[\sin(\pi) - \frac{1}{2}\sin(2\pi) \right] - \left[\sin(0) - \frac{1}{2}\sin(0) \right] = 0.
\end{aligned}$$

- By Fubini's Theorem, we also have

$$\iint_R y\sin(xy)\ \mathrm{d}A = \int_1^2 \left[\int_0^\pi y\sin(xy)\ \mathrm{d}y\right]\ \mathrm{d}x.$$

harder to integrate this way!!

- Performing integration by parts on the inner integral yields

$$\begin{aligned}
\int_0^\pi y\sin(xy)\ \mathrm{d}y &= \left[-\frac{y\cos(xy)}{x}\right]\Bigg|_{y=0}^{y=\pi} - \int_0^\pi \left[-\frac{\cos(xy)}{x}\right]\ \mathrm{d}y \\
&= \left[-\frac{y\cos(xy)}{x}\right]\Bigg|_{y=0}^{y=\pi} + \int_0^\pi \frac{\cos(xy)}{x}\ \mathrm{d}y \\
&= -\frac{\pi\cos(\pi x)}{x} + \left[\frac{\sin(xy)}{x^2}\right]\Bigg|_{y=0}^{y=\pi} \\
&= -\frac{\pi\cos(\pi x)}{x} + \frac{\sin(\pi x)}{x^2}.
\end{aligned}$$

- Performing integration by parts again, we find that

$$\begin{aligned}\int_1^2 \left[-\frac{\pi\cos(\pi x)}{x}\right] dx &= \left[-\frac{\sin(\pi x)}{x}\right]\Bigg|_1^2 - \int_1^2 \frac{\sin(\pi x)}{x^2}\, dx \\ &= \left[\frac{\sin(\pi)}{1} - \frac{\sin(2\pi)}{2}\right] - \int_1^2 \frac{\sin(\pi x)}{x^2}\, dx \\ &= 0 - \int_1^2 \frac{\sin(\pi x)}{x^2}\, dx \\ &= -\int_1^2 \frac{\sin(\pi x)}{x^2}\, dx.\end{aligned}$$

- Therefore,

$$\begin{aligned}\iint_R y\sin(xy)\, dA &= \int_1^2 \left[\int_0^\pi y\sin(xy)\, dy\right] dx \\ &= \int_1^2 \left[-\frac{\pi\cos(\pi x)}{x} + \frac{\sin(\pi x)}{x^2}\right] dx \\ &= \int_1^2 \left[-\frac{\pi\cos(\pi x)}{x}\right] dx + \int_1^2 \frac{\sin(\pi x)}{x^2}\, dx = 0.\end{aligned}$$

Double Integrals over Type I/Type II Regions in $\mathbb{R}^2$

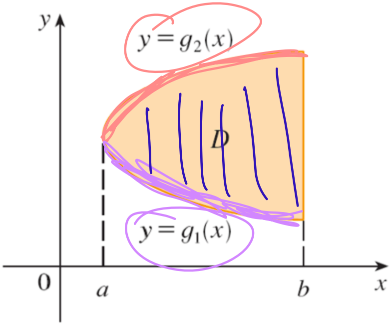
Definition (Type I Regions in $\mathbb{R}^2$)

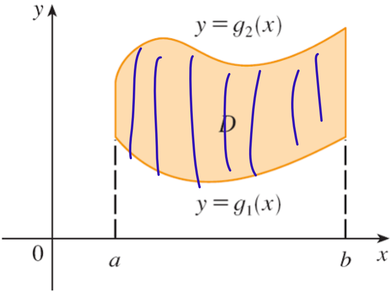
A Type I region in $\mathbb{R}^2$ is defined as a subset D of $\mathbb{R}^2$ such that there exist

- a bounded and closed interval $[a, b]$, and
- continuous real-valued functions g_1 and g_2 defined on $[a, b]$,

for which

$$D = \left\{ (x, y) \in \mathbb{R}^2 \;\middle|\; a \leq x \leq b \text{ and } g_1(x) \leq y \leq g_2(x) \right\}.$$





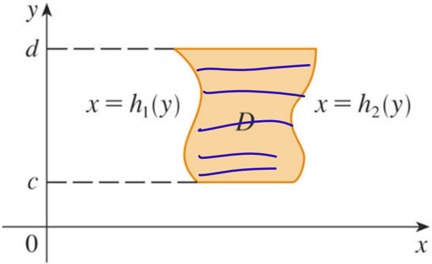
Definition (Type II Regions in $\mathbb{R}^2$)

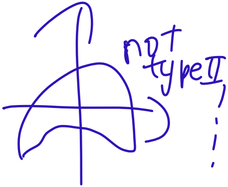
A Type II region in $\mathbb{R}^2$ is defined as a subset D of $\mathbb{R}^2$ such that there exist

- a bounded and closed interval $[c, d]$, and
- continuous real-valued functions h_1 and h_2 defined on $[c, d]$,

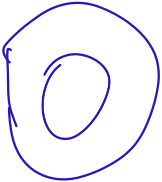
for which

$$D = \left\{ (x, y) \in \mathbb{R}^2 \;\middle|\; c \leq y \leq d \text{ and } h_1(y) \leq x \leq h_2(y) \right\}.$$

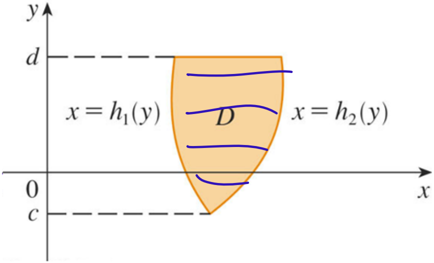




not type II, !



not type I,
not type II !!



Theorem (Double Integrals over a Type I Region in $\mathbb{R}^2$)

- Let $[a, b]$ be a bounded and closed interval.
- Let g_1 and g_2 be continuous real-valued functions defined on $[a, b]$. → always assume cont.!
- Let D denote the Type I region in $\mathbb{R}^2$ given by

$$\left\{(x, y) \in \mathbb{R}^2 \;\middle|\; a \leq x \leq b \text{ and } g_1(x) \leq y \leq g_2(x)\right\}.$$

Then

lower upper limit depend on x

$$\iint_D f(x, y)\, \mathrm{d}A = \int_a^b \left[\int_{g_1(x)}^{g_2(x)} f(x, y)\, \mathrm{d}y\right] \mathrm{d}x.$$

Remark: In the iterated double integral above, the inner integral firstly fixes $x \in [a, b]$, then integrates $f(x, y)$ with respect to y from $y = g_1(x)$ to $y = g_2(x)$.

Theorem (Double Integrals over a Type II Region in $\mathbb{R}^2$)

- Let $[c, d]$ be a bounded and closed interval.
- Let h_1 and h_2 be continuous real-valued functions defined on $[c, d]$.
- Let D denote the Type II region in $\mathbb{R}^2$ given by

$$\left\{(x, y) \in \mathbb{R}^2 \;\middle|\; c \leq y \leq d \text{ and } h_1(y) \leq x \leq h_2(y)\right\}.$$

Then

no more x left after this

$$\iint_D f(x, y)\, \mathrm{d}A = \int_c^d \left[\int_{h_1(y)}^{h_2(y)} f(x, y)\, \mathrm{d}x\right] \mathrm{d}y$$

This is y-expression

Remark: In the iterated double integral above, the inner integral firstly fixes $y \in [c, d]$, then integrates $f(x, y)$ with respect to x from $x = h_1(y)$ to $x = h_2(y)$.

Type I → y inner, x outer

Type II → x inner, y outer

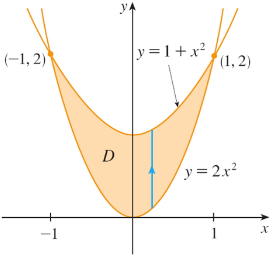
Example

Let D denote the closed region in $\mathbb{R}^2$ bounded by $y = 2x^2$ and $y = 1 + x^2$.

Problem: Evaluate $\displaystyle\iint_D (x + 2y)\, \mathrm{d}A$.

Solution

- Let us first visualize the region D.



Type I!

- D can thus be classified as a Type I region in $\mathbb{R}^2$, specified by

$$\left\{(x, y) \in \mathbb{R}^2 \;\middle|\; -1 \le x \le 1 \text{ and } 2x^2 \le y \le 1 + x^2\right\}.$$

- Then $\iint_D (x+2y)\,\mathrm{d}A = \int_{-1}^{1}\left[\int_{2x^2}^{1+x^2}(x+2y)\,\mathrm{d}y\right]\mathrm{d}x.$ upper expression lower expression
- Let us first evaluate the inner integral:

$$\begin{aligned}\int_{2x^2}^{1+x^2}(x+2y)\,\mathrm{d}y &= \left[xy+y^2\right]\Big|_{y=2x^2}^{y=1+x^2}\\ &= \left[x\left(1+x^2\right)+\left(1+x^2\right)^2\right]-\left[x\left(2x^2\right)+\left(2x^2\right)^2\right]\\ &= \left[x^4+x^3+2x^2+x+1\right]-\left[4x^4+2x^3\right]\\ &= -3x^4-x^3+2x^2+x+1.\end{aligned}$$

- We can now complete the calculation of the double integral:

$$\begin{aligned}\iint_D (x+2y)\,\mathrm{d}A &= \int_{-1}^{1}\left(-3x^4-x^3+2x^2+x+1\right)\mathrm{d}x\\ &= \left[-\frac{3x^5}{5}-\frac{x^4}{4}+\frac{2x^3}{3}+\frac{x^2}{2}+x\right]\Big|_{x=-1}^{x=1}=\frac{32}{15}.\end{aligned}$$

Practical advice for evaluating double integrals over Type I/Type II regions in $\mathbb{R}^2$

Type I region in $\mathbb{R}^2$

no gaps!

- Each vertical cross-section is a vertical line segment (or a single point).
- The vertical cross-sections are constrained between $x = a$ and $x = b$.
- For each $x \in [a, b]$, the vertical cross-section through x goes from $g_1(x)$ on the lower boundary to $g_2(x)$ on the upper boundary.

Type II region in $\mathbb{R}^2$

no gaps!

- Each horizontal cross-section is a horizontal line segment (or a single point).
- The horizontal cross-sections are constrained between $y = c$ and $y = d$.
- For each $y \in [c, d]$, the horizontal cross-section through y goes from $h_1(y)$ on the left boundary to $h_2(y)$ on the right boundary.

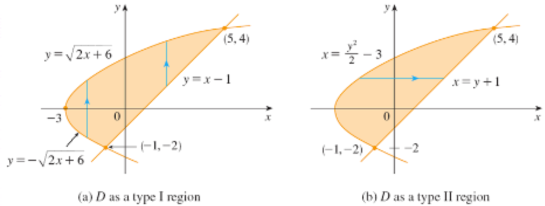
Example

Let D denote the closed region in $\mathbb{R}^2$ bounded by $y = x - 1$ and $y^2 = 2x + 6$.

Problem: Evaluate $\iint_D xy \, dA$.

Solution

- Observe that D can be classified as a Type I region or a Type II region in $\mathbb{R}^2$:



(a) D as a type I region (b) D as a type II region

- However, we prefer to work with D as a Type II region in $\mathbb{R}^2$ (think why):

$$D = \left\{ (x, y) \in \mathbb{R}^2 \;\middle|\; -2 \le y \le 4 \text{ and } \frac{y^2}{2} - 3 \le x \le y + 1 \right\}.$$

type I left curve has 2 expressions ↳ more complicated.

- Therefore,

right curve

$$\iint_D xy\ \mathrm{d}A = \int_{-2}^{4}\left[\int_{\frac{y^2}{2}-3}^{y+1} xy\ \mathrm{d}x\right]\mathrm{d}y$$

left curve

$$= \int_{-2}^{4}\left[\frac{x^2y}{2}\right]\Bigg|_{x=\frac{y^2}{2}-3}^{x=y+1}\mathrm{d}y$$

$$= \frac{1}{2}\int_{-2}^{4}\left[x^2y\right]\Big|_{x=\frac{y^2}{2}-3}^{x=y+1}\mathrm{d}y$$

$$= \frac{1}{2}\int_{-2}^{4}\left[(y+1)^2y - \left(\frac{y^2}{2}-3\right)^2 y\right]\mathrm{d}y$$

$$= \frac{1}{2}\int_{-2}^{4}\left(-\frac{y^5}{4}+4y^3+2y^2-8y\right)\mathrm{d}y$$

$$= \frac{1}{2}\left[-\frac{y^6}{24}+y^4+\frac{2y^3}{3}-4y^2\right]\Bigg|_{-2}^{4}$$

$$= \frac{1}{2}[64-(-8)] = \frac{1}{2}(72) = 36.$$

Example

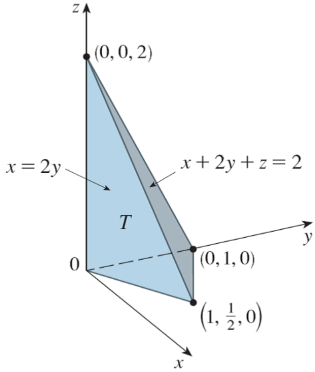
Problem: Find the volume of the closed tetrahedron T bounded by the three planes

Find x, y, z intercepts

$$x + 2y + z = 2, \quad x = 2y, \quad \text{and} \quad z = 0.$$

Solution

- Let us first visualize the closed tetrahedron T.

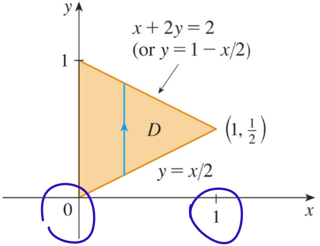


Double Integrals over Type I/Type II Regions in $\mathbb{R}^2$ — Example 3 (cont'd)

- If D denotes the Type I region in $\mathbb{R}^2$ specified by

$$\left\{(x,y) \in \mathbb{R}^2 \;\middle|\; 0 \leq x \leq 1 \text{ and } \frac{x}{2} \leq y \leq 1 - \frac{x}{2}\right\},$$

then the volume V of T is that between $D \times \{0\}$ and the plane $z = 2 - x - 2y$.



- Hence,

$$V = \iint_D \overbrace{(2 - x - 2y)}^{2}\, \mathrm{d}A = \int_0^1 \left[\int_{\frac{x}{2}}^{1-\frac{x}{2}} (2 - x - 2y)\, \mathrm{d}y\right] \mathrm{d}x.$$

- The iterated double integral can be evaluated as follows:

$$\begin{aligned}
&\int_0^1 \left[\int_{\frac{x}{2}}^{1-\frac{x}{2}} (2-x-2y)\ \mathrm{d}y\right] \mathrm{d}x \\
&= \int_0^1 \left[(2-x)y - y^2\right]\Big|_{y=\frac{x}{2}}^{y=1-\frac{x}{2}}\ \mathrm{d}x \\
&= \int_0^1 \left[\left[(2-x)\left(1-\frac{x}{2}\right) - \left(1-\frac{x}{2}\right)^2\right] - \left[(2-x)\left(\frac{x}{2}\right) - \left(\frac{x}{2}\right)^2\right]\right] \mathrm{d}x \\
&= \int_0^1 (x-1)^2\ \mathrm{d}x \\
&= \left[\frac{(x-1)^3}{3}\right]\Big|_0^1 \\
&= 0 - \left(-\frac{1}{3}\right) \\
&= \frac{1}{3}.
\end{aligned}$$